Executive Summary — Credit Card Customer Behavior Analysis

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DATASET

Source: CC GENERAL (Kaggle)
Records: 8,950 credit card customers
Variables: 18 behavioral & financial
After cleaning: 8,636 records (<4% removed)

All major financial variables are **right-skewed** (PURCHASES skew = 8.14; CASH_ADVANCE = 5.17; BALANCE = 2.39). A log-transform was applied to BALANCE before the t-test to meet normality requirements.

RESEARCH QUESTION

"Do distinct patterns of credit card usage carry statistically measurable differences, and can we identify natural customer profiles through clustering?"

Answered in two parts: **(1)** Formal hypothesis testing and **(2)** K-Means unsupervised clustering.

H1 — CONFIRMED

Claim: Heavy cash-advance users carry higher balances.

Test:	Welch's two-sample t-test (one-tailed)
t-stat:	≈ 28
df:	$\approx 2{,}889$
p-value:	$< 2.2 \times 10^{-16}$
Decision:	**Reject** H_0

Conclusion: Heavy CA users carry **significantly higher** balances. `CASH_ADVANCE_FREQUENCY` is a reliable credit-risk signal.

H2 — DEBUNKED ×

Claim: Balance strongly predicts credit limit.

Test:	Pearson correlation `cor.test()`
r:	≈ 0.54
r^2:	$\approx 29\%$ variance explained
p-value:	$< 2.2 \times 10^{-16}$
Decision:	**Debunked**

Key lesson: $p < 0.05 \neq$ practical strength. Balance explains only 29% of credit-limit variance. Banks set limits *prospectively* (income, credit history), not based on current debt.

K-MEANS CLUSTERING (K = 5)

Input: 8 standardized financial variables
Method: K-Means, `nstart=25`, `set.seed(123)`
K selected by: Elbow method + interpretability

Cluster	%
Low-Activity Customers	34%
Regular Purchasers	26%
Full-Payment Customers	18%
High CA Users	20%
High-Value Customers	1.3%

High CA Users cluster independently corroborates H1: highest CASH_ADVANCE_FREQUENCY → highest average BALANCE.

KEY CORRELATIONS

Pair	r
PURCHASES ↔ PAYMENTS	+0.60
CASH_ADVANCE ↔ BALANCE	+0.50
PRCFULLPAYMENT ↔ BALANCE	−0.32
PURCHASES ↔ ONEOFF_PUR	+0.92

TOOLS & METHODS

Language:	R
Wrangling:	`dplyr` (`filter`, `mutate`)
H1 test:	`t.test(var.equal=FALSE)`
H2 test:	`cor.test(method="pearson")`
Clustering:	`kmeans(centers=5, nstart=25)`
Transform:	`log1p(BALANCE)`

OVERALL CONCLUSIONS

- **H1 Confirmed:** Cash-advance frequency is a measurable credit-risk predictor.
- **H2 Debunked:** Statistical significance alone is misleading with large n; always report r^2.
- **Clustering:** 5 interpretable profiles reveal distinct behavioral segments; exploratory, not definitive.
- **Core lesson:** Significance $\neq$ Practical relevance. Effect size matters.